

SPRINT 01

PO/PM AGREEMENT & DECISION RECORD

Industrial Equipment Monitoring Platform

Document	Sprint 01 PO/PM Agreement and Decision Record
Milestone	M1 - Contracts and Vertical Slice
Role perspective	Product Owner / Project Manager
Requirements baseline	Version 1.0 - Draft baseline
Release	Portfolio MVP
Status	Approved for implementation
Date	19 September 2026

Purpose: freeze Sprint 01 product and delivery decisions so Firmware, Gateway, Backend and QA can implement against one deterministic telemetry contract.

Executive Approval

PO/PM decision: Sprint 01 is approved as a contract-first sprint. M1 is limited to the telemetry contract, schema/migrations, simulator-to-API ingestion, deterministic persistence, and the ordering semantics needed to prevent later redesign. Full physical integration, durable gateway reliability, and complete alert workflow remain in later milestones.

1. Agreed Sprint Goal and Scope

Sprint Goal: Establish a stable, versioned telemetry contract and prove that deterministic simulated equipment readings can travel through the ingestion API into PostgreSQL with correct identity, validation, timestamp, idempotency, partial-batch and ordering behaviour.

A successful Sprint 01 is not measured by dashboard completeness or physical hardware integration. It is measured by deterministic contract behaviour and a working simulator -> API -> persistence vertical slice that later gateway and alert work can rely on without schema redesign.

1.1 Scope commitment

Area	Sprint 01 decision
Telemetry request/response contract	In scope
Event identity and idempotency	In scope
Timestamp ownership/semantics	In scope
Database schema + migration	In scope
Simulator -> API -> PostgreSQL	In scope
Mixed/partial batch processing	In scope
Duplicate/conflict behaviour	In scope
Current-reading/replay protection	In scope
Alert ordering contract	In scope - contract only
Full alert engine	Deferred to M4
ESP32 -> MQTT -> Raspberry Pi physical path	Deferred to M2
Durable gateway buffering/retry/restart	Deferred to M3
Full authentication/site permission workflow	Deferred to M4
Performance/security/backup release evidence	Deferred to later milestones

2. Approved Decisions

PO-D01 - Canonical event identity

Status: APPROVED

The canonical telemetry identity is:

(device_id, boot_id, sequence_number)

The database must enforce:

```
UNIQUE(device_id, boot_id, sequence_number)
```

Retries retain the same three values. A reboot creates a new boot_id; sequence_number may restart within that new boot.

Existing data	Incoming data	Backend result
No existing identity	New event	accepted
Same identity + same immutable telemetry	Retry	duplicate
Same identity + changed immutable telemetry	Identity reuse	rejected: identity_conflict

Duplicate comparison for the MVP uses immutable event fields directly: schema_version, device_id, boot_id, sequence_number, measured_at, device_uptime_ms, metric, value, unit and quality. Gateway/backend receipt timestamps are not immutable device content.

PO/PM conclusion: QA-OQ05 is closed. A canonical content hash may be used internally later, but it must not change contract semantics.

PO-D02 - Timestamp ownership

Status: APPROVED

Field	Owner	Meaning
measured_at	Device	Physical measurement time when the device clock is trustworthy
device_uptime_ms	Device	Elapsed time since the current device boot
gateway_received_at	Gateway	Original time the gateway received the event
backend_received_at	Backend	Backend receipt/processing time

All persisted absolute timestamps use UTC. backend_received_at is backend-owned and must never be supplied by a client. gateway_received_at represents original gateway receipt, not later replay time.

Critical rule: measured_at is never fabricated. If the device clock is not trustworthy, measured_at is null; the system must not silently substitute gateway_received_at or backend_received_at.

PO-D03 - Minimum clock-quality model

Status: APPROVED for MVP

Sprint 01 supports the minimum clock vocabulary:

```
quality.clock:
  synchronised
  unsynchronised
```

```
synchronised
  -> measured_at may be trusted
```

```
unsynchronised
  -> measured_at must be null
  -> gateway_received_at becomes the fallback timeline
```

quality.reading remains logically separate from clock quality. Estimated time, NTP metadata, uncertainty ranges and richer clock states are deferred unless implementation evidence requires them.

3. Partial Batch Outcomes and Queue Behaviour

PO-D04 - Mixed batches are not application-level all-or-nothing

Status: APPROVED

Every event receives an independent deterministic result: accepted, duplicate, or rejected. One invalid telemetry item must not block unrelated valid events.

```
101 -> accepted
102 -> duplicate
103 -> rejected: invalid_unit
104 -> accepted
```

Backend outcome	Gateway action
accepted	Delete from queue
duplicate	Delete from queue
rejected + permanent reason	Move to bounded quarantine and remove from delivery queue
timeout / 429 / 5xx / transient backend failure	Keep queued and retry using bounded backoff

PO-D05 - Database commit boundary

Status: APPROVED / NON-NEGOTIABLE

The backend must never report an item as accepted before the corresponding PostgreSQL transaction has committed. If the backend commits successfully but the response is lost, the gateway retries the same identity and receives duplicate rather than creating a second row.

PO-D06 - HTTP semantics for mixed batches

Status: APPROVED

A syntactically valid batch that is successfully processed returns 200 OK even when its individual items contain a mixture of accepted, duplicate and rejected outcomes. The per-item result is authoritative.

Condition	Request-level response
Valid batch; mixed item outcomes	200 OK
Malformed/envelope-invalid request	400/422
Authentication/authorisation failure	401/403
Batch exceeds configured maximum	413
Rate limited	429
Database/service transient failure before committed result	5xx; no uncommitted accepted claims

The maximum batch size remains 500 readings.

4. Rejection Taxonomy

PO-D07 - Machine-readable rejection reasons

Status: APPROVED for contract v1

The initial permanent rejection vocabulary is:

```

unknown_device
wrong_gateway
unsupported_schema_version
invalid_metric
invalid_unit
value_out_of_range
malformed_value
identity_conflict

```

Gateway logic must branch on stable machine-readable codes, never on human-readable error text. QA-OQ07 is closed for API v1.

5. Alert Ordering Contract

The full alert engine remains deferred, but ordering semantics are frozen now because persistence, replay and current-reading behaviour depend on them. Baseline behaviour to preserve includes three consecutive valid high readings to open an alert, three recovery readings to resolve it, a >15-second gap reset, exclusion of invalid readings, historical-only handling for readings more than 30 seconds old at backend receipt, and no live advancement from duplicates or out-of-order events.

PO-D08 - Within-boot ordering

Status: APPROVED with clarification

For events in the same (device_id, boot_id), sequence_number is the canonical progression order. An event with a sequence number lower than the highest live-evaluated sequence is persisted historically but cannot rewind current state or advance live alert state.

```
last live evaluated sequence = 103
```

```

incoming seq 101
-> persist in history
-> DO NOT rewind current reading
-> DO NOT advance alert state

```

Sequence numbers do not need to be contiguous. Missing sequence numbers do not themselves reset a pending alert count; the time-gap rule controls continuity.

PO-D09 - Effective event time

Status: APPROVED

```

effective_event_time =
  measured_at      if quality.clock == synchronised
  gateway_received_at otherwise

```

sequence_number is the logical ordering gate within a boot. effective_event_time is used for age and gap calculations. These are separate concepts and must not be conflated.

PO-D10 - 30-second stale rule

Status: APPROVED

```

backend_received_at - effective_event_time > 30 seconds
-> store as history
-> DO NOT advance live temperature alert state
-> DO NOT replace newer current state

```

QA-OQ03 is closed.

PO-D11 - 15-second consecutive-reading gap

Status: APPROVED

```
current.effective_event_time
- previous_live_evaluated.effective_event_time
> 15 seconds
-> reset trigger/recovery consecutive-reading counter
```

QA-OQ04 is closed.

6. Cross-Boot Ordering

PO-D12 - Reboot establishes a new continuity boundary

Status: APPROVED

Sequence numbers cannot establish recency across two different boots. For the MVP, the backend may derive an estimated boot epoch as follows:

```
estimated_boot_start = gateway_received_at - device_uptime_ms
```

The current live boot is the boot with the latest established boot-start epoch. Once boot-B has become current, delayed boot-A telemetry may populate history but must not replace current temperature or advance live alerts.

For the MVP, gateway_received_at is accepted as authoritative enough for this ordering assumption. Stronger gateway-clock integrity can be added later.

PO-D13 - Reboot resets pending alert continuity

Status: APPROVED

A reboot resets an incomplete HIGH_PENDING or RECOVERY_PENDING counter. Readings from different boots are not combined to satisfy a three-consecutive-reading condition.

7. Alert Rule Scope

PO-D14 - MVP rule scope

Status: APPROVED

For the Portfolio MVP, use one organisation-level temperature rule configuration covering trigger threshold, trigger consecutive count, recovery threshold and recovery consecutive count. It applies to temperature-enabled devices. Per-site, per-device and device-class overrides are deferred.

8. Remaining Open Questions

ID	Question	Status	Owner	Sprint impact
OQ-01	Exact modelling of a disconnected/invalid sensor reading versus telemetry value validation	Open	Firmware + Backend + QA	Resolve before M2
OQ-02	Exact bounded quarantine capacity and retention	Open	Gateway Lead + Tech Lead	M3; non-blocking for M1 contract
OQ-03	Strategy for concurrent batches affecting the same device alert state	Open	Backend Lead	Must close before M4 alert implementation
OQ-04	Whether future clock quality needs estimated/uncertainty/NTP state	Deferred	Tech Lead	Not required for MVP
OQ-05	Gateway clock-drift monitoring/tolerance	Deferred	Gateway Lead	Assumption documented

ID	Question	Status	Owner	Sprint impact
				for MVP

9. PO/PM Sprint 01 Exit Position

Sprint Goal and scope are agreed. Sprint 01 can be signed off when the versioned telemetry contract, database uniqueness migration, mixed-batch API, simulator vertical slice and automated tests demonstrate the approved decisions in this record.

Required evidence includes valid, duplicate, conflicting-identity, mixed-batch, lost-response/retry, stale, replay and out-of-order scenarios, with deterministic results that Firmware, Gateway, Backend and QA can independently predict from the same contract.

Project status after PO/PM review: M1 scope approved. Core event identity, timestamp, partial-batch and alert-ordering semantics are sufficiently defined for implementation. Remaining questions are either assigned to later milestones or must be resolved before dependent work begins.

10. Source Documents

- Industrial Equipment Monitoring Platform - Requirements Specification, Version 1.0 Draft baseline, especially Sections 5-7 and 11-12.
- QA Sprint 1 - Telemetry Contract, Decision Log and Open Questions, especially Sections 1-8.
- Sprint 01 Technical Report - Technical Lead Report, M1 Contracts and Vertical Slice, especially Sections 1-11 and Appendix A.

This document consolidates and approves the PO/PM decisions derived from those project sources. It does not replace the baseline requirements specification; changes to the baseline must continue to record affected requirement IDs, rationale, acceptance-test changes and schedule impact.